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Overcoming Geological Unconformities and Laterally Variable Lithology for Optimal Well Placement

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Abstract

The development of carbonate fields poses significant challenges due to complex geological structures, unconformities, and lateral variability in reservoir properties. The Upper Shuaiba formation in northern Oman has stratigraphic reservoirs with argillaceous units and clean carbonates units with a varying thickness between 1 and 6 meters. Geological understanding of facies distribution within upper Shuaiba formation is key to assess reservoir quality. Well placement within the Upper Shuaiba clinoform presents significant challenges due to the thin and variable reservoir thickness. Initial attempts using conventional methods often resulted in suboptimal positioning and variable reservoir performance, highlighting the need for advanced geosteering technologies. Geological studies highlight the importance of understanding lateral facies variations within the Upper Shuaiba formation for assessing reservoir quality.

This study explores the application of high-definition deep resistivity mapping tools to enhance well placement and optimize reservoir management in such challenging environments. By accurately visualizing subsurface boundaries based on resistivity contrasts, these advanced technologies mitigate the risks associated with sudden exits and poorly defined reservoir limits. Pre-well modelling done based on only one offset well indicated good contrast between both upper and lower boundaries. In addition, density image, and porosity tools were deployed to provide a detailed picture of the reservoir and refine the geological interpretation.

Resistivity variation across the formation boundaries was approximately 1 to 8 Ohmm and the total thickness of the target zone based on offset data indicated 1.9m. However, there was expectation of possible thinning interpreted from previous well exits in this formation. In the only nearby offset well, well placement was optimized based on conventional deep azimuthal EM resistivity without imaging capabilities. In this well a sudden exit occurred without any reasonable anticipation which led to moving the well azimuth far to the east side to increase reservoir exposure. The expectation was that the reservoir was thinning or truncated towards the west side. In the planning stage of the target well, the asset team decided to locate a new well plan towards the West side of the previous well to eliminate any doubt regarding thinning or erosion of the reservoir and inspect a new tools capability. The resistivity imaging allowed boundaries to be clearly identified and mapped the true thickness of the target zone, allowing optimization of the well placement position in the best reservoir zone.

The implementation of these advanced geosteering technologies has proven transformative, enabling operators to overcome the geological challenges associated with unconformities and lateral resistivity changes while minimizing operational risks and costs. The new generation EM tools with enhanced sensitivity proved to be capable of boundary tracking, enabling high confidence in well placement operations, allowing accurate geosteering and improved reservoir understanding.

Introduction

This study focuses on an area in northern Oman within the Upper Shuaiba clinoforms, which are geological formations made up of complex layers called clinoforms. These clinoforms are important for understanding the local rock layers because they show changes in how sediments were deposited and what kinds of rocks formed. The Shuaiba Reservoir is truncated by the Nahr Umr shale, which adds to the complexity and makes exploration and production more challenging.

Previous geological studies have given us important information about the Upper Shuaiba formation. It mainly consists of thin carbonate clinoforms, usually between 1 and 6 meters thick, separated by clay-rich layers. These clinoforms formed as carbonate sediments built out toward the Bab Basin in the UAE. This process affected how reservoir rocks are spread out across the field (Figure-1).

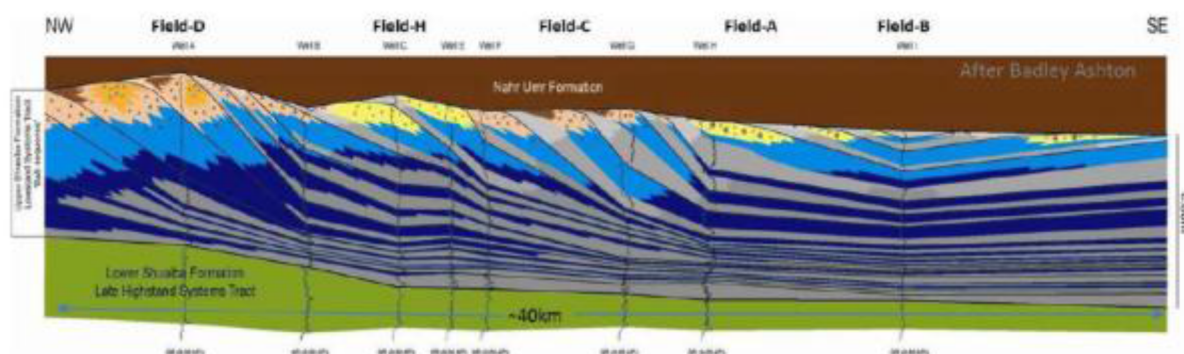


Figure 1—NW-SE geological cross and associated regional depositional model of the Upper Shuaiba clinoforms as part of the infill in the late Aptian-aged Bab Basin. (Al Badi et al 2024)

The clinoforms are labeled A to K, from youngest to oldest, and show different rock types depending on water depth and sea level when they formed. As you go from shallow to deeper water, the rocks and their qualities change.

The best reservoir rocks are those with rudists and grainstone, which have good porosity and permeability. Rudist are fossil shells of creatures that lived in shallow, warm water and helped form these rocks. Grainstone is made of grains cemented together, usually found in high-energy shallow areas like shoals. On the other hand, microporous and cemented grainstones have poor reservoir qualities and are found in deeper water. Each clinoform truncated at the Nahr Umr formation, with thick clay-rich layers in between that act as barriers, stopping fluid flow between clinoforms and keeping reservoirs separate.

Knowing where the good and poor reservoir rocks are, and how the layers stack, helps us find and produce oil and gas more effectively. This study focuses on the Upper Shuaiba Clinoform B (USB), which has two main parts: B1 and B2. B1 has better reservoir properties, while B2 is denser and tighter. Although drilling mainly targets B1, its thinness and well placement issues sometimes require drilling into both B1 and B2.

In the area studied, two rock types—rudist and microporous—are found and produce hydrocarbons. There is only one well with data on the USB reservoir (rudist type) in this area, all drilled in the subcrop part of USB. Based on reservoir data logs, the USB reservoir here is mostly made of rudist rock. The relatively low oil saturation in this well is probably due to well placement (Al Badi et al., 2024).

Understanding the detailed geological characteristics of the USB is critical for improving hydrocarbon recovery. Factors such as variations in rock porosity, permeability, and fluid saturation influence how oil and gas move through the reservoir. This study aims to better characterize these properties and improve predictions of reservoir behavior during production.

Moreover, the presence of clay-rich barriers between clinoforms means that each reservoir unit can act independently. This compartmentalization affects pressure communication and fluid flow, which has direct implications for well design and field development strategies. By studying these barriers and reservoir quality variations, operators can optimize well placement, reduce drilling risks, and improve overall production efficiency.

Finally, the study contributes to the broader understanding of carbonate clinoform reservoirs, which are common in many hydrocarbon-producing regions worldwide. The knowledge gained here can be applied to similar geological settings, helping to advance exploration and production technologies.

Well placement technology advancement

Advances in LWD electro-magnetic (EM) deep resistivity tools design

Azimuthal Deep resistivity sensors. Bittar et al. (2007) describe how a deep resistivity LWD tool design utilizing tilted receiver antenna helped in geo-steering and formation evaluation. Figure-2 shows a diagram of the described azimuthal deep resistivity tool which was used in old wells. While the transmitters were all co-axial, receivers (in red color) were tilted. The two receivers in the middle provided multi-spacing compensated resistivity measurements with azimuthal sensitivity, these measurements are used for a petrophysical analysis and with the azimuthal sensitivity also allows geosteering experts to identify the direction of resistivity changes around the wellbore. A single receiver was also added at the bottom the tool (deep receiver) to provide a wider transmitter to receiver spacing to enable signals to be fired deeper into the formation (illustrated through dashed lines). While a single receiver wouldn't provide an absolute resistivity value a so-called geo-signal output from it detects resistivity boundaries around the wellbore. Depending on the resistivity environment around the wellbore the tool was designed to achieve up to 15ft + depth of investigation (DOI) for bed boundary mapping around the tool using the furthest transmitter-receiver spacing and lowest frequency.

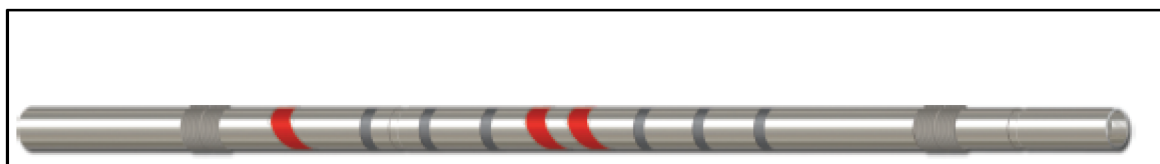


Figure 2—Older deep resistivity tool design

New Generation Deep Resistivity Sensors. Wu et al (2022) describes a new design of deep resistivity EM tools. The newer deep resistivity tool as shown in Figure-3 which was used in this study provides several improvements over the older design in terms of:

- Longer spacing & lower frequency: the largest transmitter-receiver spacing is now 12ft and tilted deep transmitters can fire EM signals down to 62.5kHz enabling deeper travel into the formation. This is now enabled because of the higher signal strength and signal-to-noise ratio (SNR) provided by the electronics.
- Crossed Transmitter: providing near wellbore R_v / R_h (vertical and horizontal resistivities) and formation dip.

- Tilted transmitters and receivers: while tilted receivers in the older design provided the basic need to have azimuthal sensitivity to changes in resistivity around the wellbore, utilizing the tilted antenna design on both transmitters and receivers means much more available EM signal components that can be utilized into a high-resolution deep resistivity inversion.

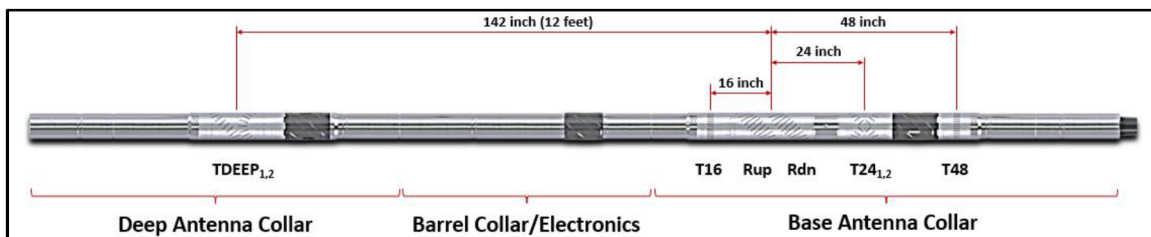


Figure 3—Newer deep resistivity tool design

One of the offset wells was drilled using the older generation of deep resistivity tool [Figure-2](#), while in this case study was drilled using the new high-definition deep resistivity tool providing multi-layer inversion which is optimum for such complex geology [Figure-3](#).

Challenges in Well Placement

Each clinoform varies in thickness and reservoir properties, with the sweet zone typically averaging between 1m and 4m in thickness. Porosity ranges from 15% to 22%, with relatively low permeability varying from 0.1 to several millidarcies (mD). All clinoforms show low gamma-ray (GR) readings of 10-20, and density values range from 2.1 to 2.4. Nonetheless, resistivity values show greater variation, with averages exceeding 10 ohmm in rudist and grainstone units, while values in microporous rocks range from 1-2 ohm. ([Alsulaimani et al 2025](#))

Only one pilot hole was drilled in the AoI, and a sidetrack was performed to orient horizontally inside the reservoir ([Figure-4](#)). The results of the pilot hole indicated good porosity and saturation in the B1 unit, with a thickness of 1.9 meters. However, during the drilling of the lateral section, the average reservoir properties were much lower than those found in the pilot hole. Attempts were made to adjust the azimuth to achieve similar properties to the pilot hole, but the average results remained low.

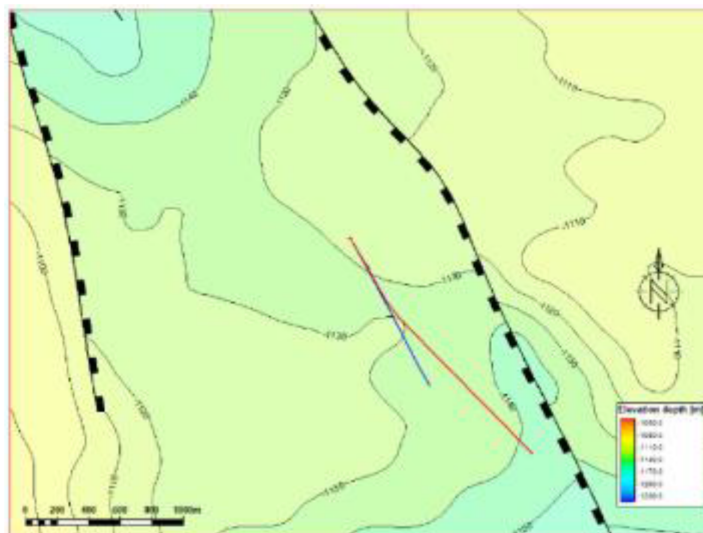


Figure 4—The first well drilled in the polygon aimed to locate best reservoir properties, starting along the blue line, and adjusting azimuth as indicated by the red line. ([Al Badi et al 2024](#))

The case study well targeting the Rudist rock type in B clinoform. It was drilled from the NW to the SE. The NW-SE orientation of the well pattern assures to reduce the risk of intersecting sub-seismic faults which causes exiting from the reservoir (orientation parallel to the dominant Alpine NW-SE fault set) and parallel to the existing well.

Analysis of Offset Well

The main objective of the geosteering is to maximize exposure in an interval of thin reservoir in order to tap reserves at an economic rate. The vertical thickness of the target zone was estimated to be around 1.9m with lateral dip variation. Based on offset wells, presence of fracture corridors was expected. Moreover, formation thinning was anticipated with properties deteriorating towards the truncation line. Hence, proactive geosteering would be always required to place the well within the target zone. Geosteering model is created based on the closest available near-vertical offset well as shown in Figure-5.

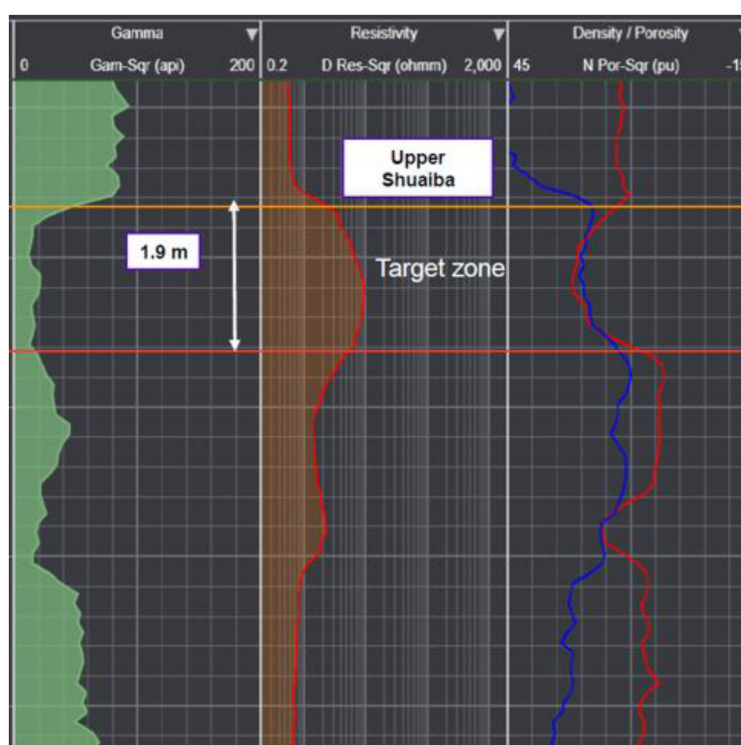


Figure 5—Closest available offset well used to create Geosteering model

According to the vertical offset well, located towards the east direction, B-unit is approximately 1.9m thick with good petrophysical properties. Offset wells indicate potential thinning of the target layer as seen from the second offset well that has been geosteered using older deep resistivity generation. Three reservoir exits were encountered while drilling the offset horizontal well as shown in Figure-6. General reservoir properties showed GR of 11-20 api, resistivity ranging from 6 to 10 ohm-m with lower resistivity zones reaching 2 ohm-m in some intervals which correlates to the porosity readings. Density at landing was averaging 2.3 g/cc for first 50m then remarkably deteriorated till the well TD.

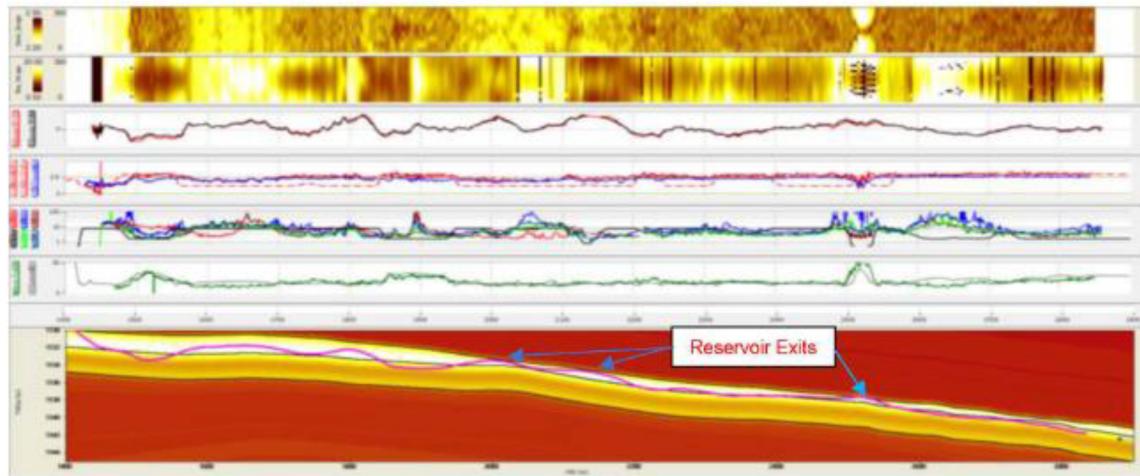


Figure 6—Geosteering in this field using the previous generation of the azimuthal resistivity was always challenging and ending up with multi exists. (Alsulaimani et al 2025)

Pre-well Model

Based on the pre-well modeling, it was decided to place the deep resistivity tool close to the bit as it is expected to provide key information for geosteering decisions in real-time. Placing this tool as close to the bit as possible allowed early detection of resistivity changes with respect to the boundary position which had contrast in the resistivity readings.

In addition to pre-well modelling, the only offset horizontal well has also been evaluated. Figure-7 and 8 shows the offset well to define the challenge and optimum well placement strategy. It is important to consider that these horizontal wells show formation heterogeneity which could be overlooked if the planning is primarily based on vertical offset wells. Where lateral formation properties change and dip variation is evident, identifying that this can happen in a target reservoir can minimize potential risks when considered for the lateral section rather than relying only on vertical pilot holes.

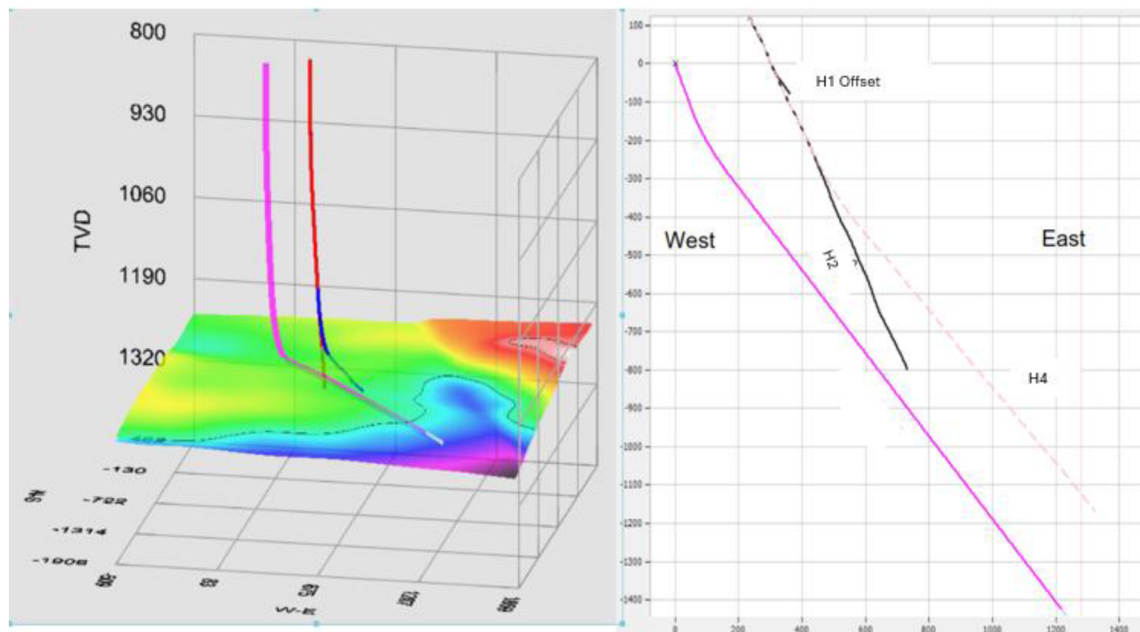


Figure 7—Location of planned well with reference to offset well. (Magenta is the planned well)

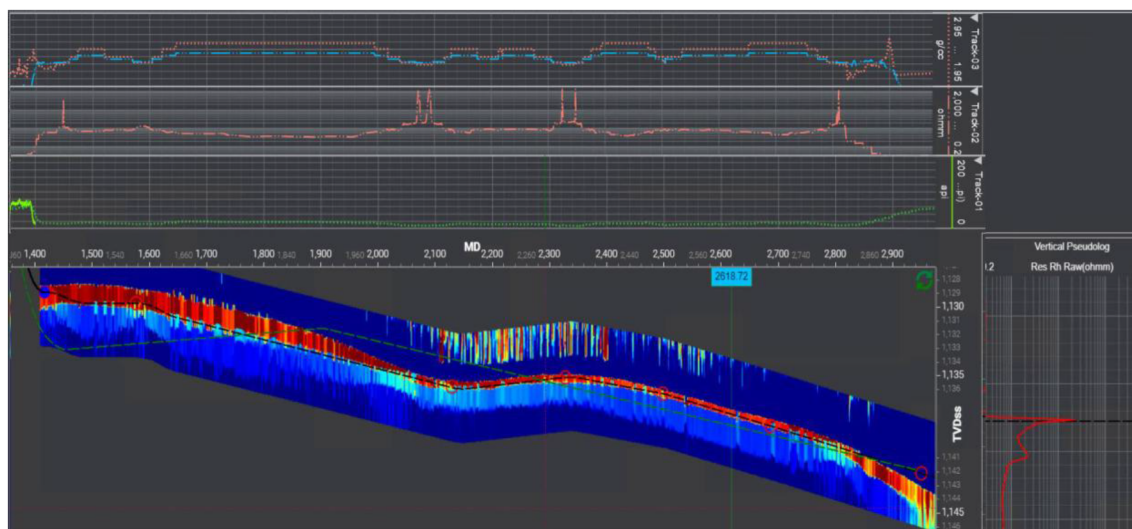


Figure 8—Re-modelling of offset well H2 using high-definition resistivity inversion

In the same area of offset well; slanted 8 ½" vertical hole was drilled to provide a control point, helping to tag reservoir for H2 and mitigate a risk of reservoir exit. While the vertical hole shows a ~1.9m thick B-reservoir with good characterizations, deviated well showed approximately 5m at landing. After landing at offset well H2; the lateral section was stepping towards west till exiting the B-unit. At exit the thickness of B-unit was ~0.5m indicating target zone thinning towards the west direction.

As a contingency plan, H3 plan prepared with azimuth correction towards the East from H2 and continued drilling till exit target zone trying to search for better reservoir properties. H4 plan prepared at the same direction as H3 and drilled to ~2450m then appraised the base part till 2775m at 81°

At the current well, the trajectory is stepping more to west where there was expected more thinning as per the H2 offset well. The target zone exposed to an unconformity surface might lead to a sudden exit of thin target zone. Any sudden dip change could lead to an exit due to it being hard to maneuver in the thin unit.

Results

The well placement was quite challenging due to inconsistent thickness, dip facies changes, lateral property variations and heterogeneity of the reservoir itself. However, using the new tool in conjunction with other logs helped to understand the reservoir and place the wellbore within the best performing intervals of the formation. The well was drilled from casing shoe to TD with a total lateral length of 6 1/8" well section of 1827m MD, which was all drilled within the target reservoir of USB clinoform.

The new resistivity tool was able to map the target B-unit along the lateral section, revealing complex structure of target USB reservoir and its resistivity profile. The inversion allowed to define confidently thicknesses, dips and well position relative to boundaries, confirmed by conventional logs. Overall, by using inversion results, lower resistivity intervals were confirmed withing the target reservoir.

The study, the formation had a stratigraphically heterogeneous resistivity profile in some intervals, that could be misleading in case of definition of the well positioning using geosignals. Thus, inversion showed beneficial performance in such intervals mapping confidently top & bottom of the sweet zone.

Figure 9 shows the well drilled in an area with significant subsurface uncertainty since there was only one nearby offset well. The objective was to geosteer using High-Definition deep resistivity inversion tool to maximize the horizontal section length to truncation line avoiding any exits close to the truncation. The well was successfully placed in the best zone throughout the horizontal section, resulting in better properties than prognosed. Additionally, the TD was maximized to the truncation line using this tool without exiting out of the sweet zone.

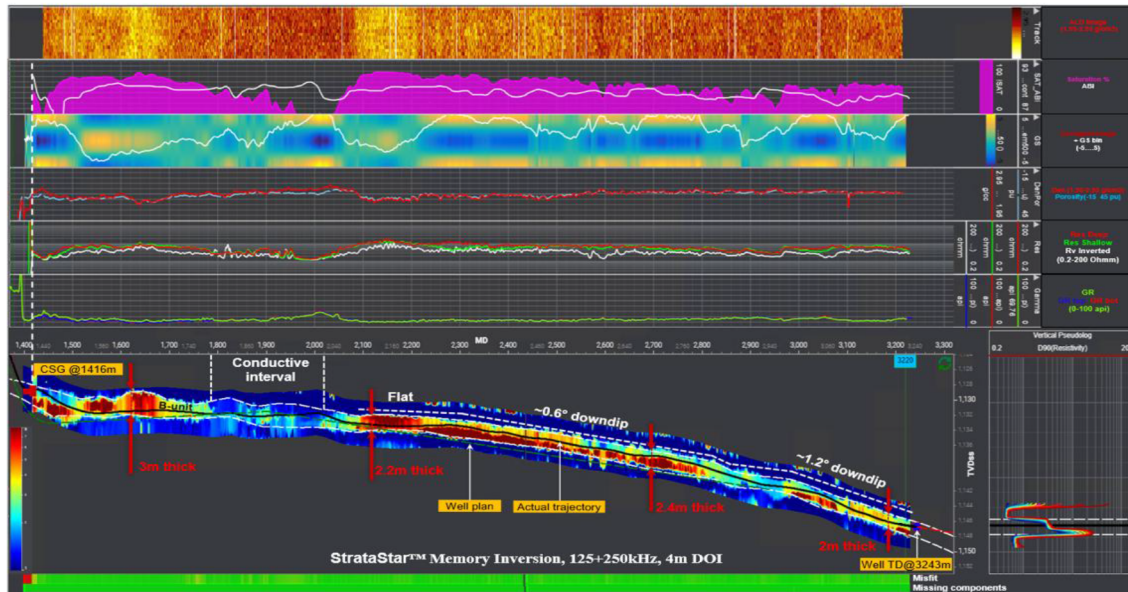


Figure 9—Final result of the new advanced generation of azimuthal resistivity-inversion tool in the well. Despite there is only one offset well that controls the structure, the tool was able to map the boundaries up to the truncation line.

Conclusion

This case study illustrates how these technologies have been instrumental in optimizing well placement within the Upper Shuaiba formation. Through accurate mapping of reservoir boundaries and lithological variations, geoscientists and engineers can now make informed decisions that mitigate drilling risks and maximize reservoir contact. This capability not only improves production rates and recovery efficiencies but also reduces operational costs and enhances overall project economics.

Based on the successful outcome, plans are now underway to further develop the entire polygonal area. More than 10 pairs of wells, comprising both oil producers and water injectors, are scheduled for drilling. These wells will be strategically spaced at an average interval of 200 meters, covering approximately 1500 meters in lateral extent. This comprehensive development plan aims to maximize reservoir recovery efficiency while minimizing operational risks and costs. (Al Badi et al 2024)

In conclusion, the integration of advanced geosteering technologies marks a significant milestone in the evolution of reservoir management. By enabling precise well placement and enhanced reservoir characterization, these tools empower operators to achieve superior outcomes in hydrocarbon recovery. As the industry increasingly adopts new technologies, geosteering will remain crucial in shaping the future of energy exploration and production. It will drive improvements in efficiency, sustainability, and economic success in the global energy sector.

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